



Interventional antibiotic treatment replacing antibiotic prophylaxis during allogeneic hematopoietic stem cell transplantation is safe and leads to a reduction of antibiotic administration

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Abstract

Patients undergoing allogeneic hematopoietic stem cell transplantation (alloHSCT) face an elevated risk of infection-related mortality, particularly during the pre-engraftment period. Although systemic antibiotic prophylaxis (SAP) is commonly employed during neutropenia, it is linked to disruptions in the intestinal microbiome, increasing the risk of graft-versus-host disease (GVHD), *Clostridium difficile* infection (CDI), and colonization with multi-drug resistant (MDR) bacteria. In our retrospective analysis, we evaluated the safety and efficacy of an exclusively interventional antibiotic treatment (IAT) compared to SAP in adult alloHSCT patients. In comparison to SAP, IAT resulted in a significantly reduced duration of antibiotic therapy (24 vs. 18 days, $p < 0.001$), although the cumulative incidence (CI) of bloodstream infections (BSI) by day + 100 post-HSCT was significantly higher in the IAT group compared to SAP (40% vs. 13%, $p < 0.001$). However, this did not lead to a significant increase in ICU transfers (13% vs. 6%, $p = \text{ns}$) or a higher CI of non-relapse mortality (NRM) at 3 years (11% vs. 10%, $p = \text{ns}$). With a median follow-up of 1052 days, the 3-year overall survival (OS) rates were 69% and 66% for the SAP and IAT cohorts, respectively ($p = \text{ns}$). The CI of acute GVHD grade II–IV (30% vs. 39%) at 100 days or chronic GVHD of any grade (50% vs. 45%) at 3 years did not differ significantly between the SAP and IAT groups. There was a tendency towards a higher CI of severe chronic GVHD in the SAP cohort (28% vs. 13%, $p = 0.08$). Our single center experience in conducting alloHSCT without antibiotic prophylaxis but with stringent guidelines for prompt antibiotic intervention demonstrated no disadvantages in terms of OS and NRM. IAT led to significantly reduced consumption of cefotaxime, carbapenem, and glycopeptide antibiotics. In conclusion, our findings suggest that replacing SAP with the proposed IAT procedure is both safe and feasible.

Keywords Allogeneic stem cell transplantation · Chronic GVHD · Acute GVHD · Antibiotic prophylaxis · Multi drug resistant bacteria · Blood stream infections

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Introduction

Allogeneic hematopoietic stem cell transplantation (alloHSCT) is a potentially curative treatment for various hematologic malignancies [1]. Despite the optimization of conditioning regimens and supportive care, alloHSCT still carries a significant risk for infection-associated morbidity and mortality [2]. Therapy-induced mucosal damage with concurrent neutropenia during the pre-engraftment period facilitates bacterial translocation into and spread in the bloodstream. These blood stream infections (BSI) affect 30–40% of alloHSCT patients being the most frequent infectious complication [3, 4].

To reduce the risk of BSI, systemic antibiotic prophylaxis (SAP e.g., ciprofloxacin, levofloxacin) covering the period of neutropenia is a widely accepted procedure for HSCT patients and dates back to a randomized trial demonstrating a reduction of Gram-negative sepsis episodes, but no significant changes in survival [5] [6]. Nevertheless, the clinical application of SAP is becoming increasingly debated. First, despite SAP administration most patients develop neutropenic fever regardless and subsequently require additional antibiotics [7]. Second, the widespread use of antibiotics (in the prophylactic setting) is associated with a strong rise in difficult-to-treat multidrug-resistant organisms (MDRO) leading to higher mortality rates [8]. Third, the use of broad-spectrum antibiotics can reduce intestinal microbiome diversity which is associated with development of GVHD [9], and inferior overall survival (OS) [9] [10]. Thereby, the microbiota composition is affected not only by the administered antibiotic regimen, but also by its timing of initiation and duration [11, 12].

Based on the increasing evidence highlighting negative associations of SAP in the context of alloHSCT, we hypothesized that an interventional antibiotic treatment (IAT) strategy is non-inferior to SAP regarding the incidence of severe infections with a simultaneous reduction of antibiotic administration, and might improve survival outcomes of patients undergoing allo HSCT. We therefore transitioned from routine application of SAP in patients undergoing alloHSCT at our centre to a SAP-free practice using interventional antibiotic treatment (IAT) in case of neutropenic fever or any other signs of infection. This retrospective analysis aimed to address both safety and efficacy of IAT in alloHSCT patients by analysing the outcome in terms of overall survival (OS), non-relapse mortality (NRM), antibiotic consumption and incidence of BSI.

Methods

Patient characteristics

We retrospectively analysed all consecutively treated adult patients undergoing alloHSCT at in the University Hospital Frankfurt, Germany. Inclusion criteria were (1) hematologic disease requiring alloHSCT, (2) age ≥ 18 years and (3) receipt of non-T cell depleted graft. Patients who underwent previous alloHSCT were excluded. The SAP cohort consisted of 67 patients who received alloHSCT in 2017, and the IAT cohort of 68 patients transplanted in 2019. Clinical metadata were obtained with institutional review board approval of the UCT and the Ethical Committee at the University Hospital Frankfurt (project-number: UCT-14-2022, SHN-1-2022).

Definitions

Conditioning regimen for alloHSCT were defined as myeloablative (MAC) or reduced intensity (RIC) according to Bacigalupo et al. and Giralt et al [13, 14]. The Hematopoietic cell transplantation specific comorbidity index (HCT-CI) was calculated to assess pretransplant comorbidities [15]. The disease risk index (DRI) according to Armand et al. was applied to facilitate comparison of patients with various hematologic diseases and disease stages [16]. Neutropenia was defined as an absolute neutrophil count of < 500 cells/ μL and neutropenic fever as an elevated body temperature $> 38^\circ\text{C}$. During each fever episode blood samples were obtained for microbiological cultures into culture bottles (BD BACTEC Lytic/10 Anaerobic/F and BD BACTEC Plus Aerobic/F; Becton Dickinson, Heidelberg, Germany) and isolated bacteria were identified with standard methods. BSI was defined as one or more positive blood culture requiring treatment with systemic antibiotics. For coagulase-negative staphylococci at least two positive blood culture results on the same day or on consecutive days were required [17]. Upon admission every patient underwent MDRO screening weekly thought their hospitalization for alloHSCT by pharyngeal, rectal, and nasal swabs. Colonization was determined by the detection of an MDRO in at least one of these swabs. In our study MDRO was defined as methicillin-resistant *Staphylococcus aureus* (MRSA), vancomycin-resistant *Enterococcus* (VRE), extended-spectrum β -lactamase (ESBL)-producing *Enterobacteriaceae*, and carbapenemase-producing *Enterobacteriaceae* [18]. Chronic GVHD (cGVHD) and acute GVHD (aGVHD) were assessed using standard definitions and stratification guidelines [19, 20].

Detection of MDRO and analyses of resistance

All laboratory procedures adhered to ISO 15189:2007 standards (certificate no. D-ML-13102-01- 00, valid through January 25, 2021). Rectal swaps were collected using Amies media and streaked onto selective agar plates (e.g., CHRO-Magar ESBL plates; Mast Diagnostica, Paris, France and ChromID VRE agar; bio- Merieux, Nurtingen, Germany). For MRSA detection, nasal swaps were streaked onto selective MRSA agar plates (e.g., Brilliance MRSA Agar; Oxoid, Wesel, Germany).

Species were identified vis biochemical identification methods or MALDI-TOF analysis. Antibiotic susceptibility was tested following CLSI guidelines using VITEK 2 and/or antibiotic gradient tests (bioMerieux) along with agar diffusion. Gram-negative, carbapenem-resistant isolates, underwent PCR-analyses to detect genes that encode carbapenemases, including blaNDM, blaVIM, blaIMP, bla-OXA-48-like, blaKPC and OXA-23, OXA-24, and OXA-58 for *A. baumannii* [21].

Cohort description

AlloHSCT procedures were performed in a dedicated transplantation unit according to local protocols. From 5 days before alloHSCT until neutrophil recovery all patients were kept in a protective environment including high-efficiency particulate air (HEPA) filtered clean air and low-microbial diet. All patients received an anti-infective prophylaxis consisting of aciclovir, cotrimoxazole, an echinocandin or broad-spectrum azole.

Cefotaxime served as the standard of care for SAP during neutropenia post-allogeneic alloHSCT until December 2017. It was administered intravenously at a dosage of 2 g twice a day, commencing approximately five days before alloHSCT, concurrently with the administration of anti-thymocyte globulin (ATG) or upon the onset of neutropenia. This regimen continued until neutrophil recovery, approximately 14 days later. In the event of the first episode of neutropenic fever, cefotaxime was replaced with piperacillin/tazobactam (4 g/0.5 g; 4 doses/day) or a carbapenem (imipenem 0.5 g; 4 doses/day or meropenem 2 g; 3 doses/day). Depending on the likelihood of a severe infection (sepsis), mucosal damage, or evidence of multidrug-resistant organism (MDRO) colonization, the antibiotic regimen was escalated with a glycopeptide (vancomycin or teicoplanin, dosed according to the patient's weight and blood concentration).

Patients in the IAT cohort were not treated with prophylactic cefotaxime. In the event of the initial occurrence of neutropenic fever patients received either piperacillin/tazobactam (4 g/0.5 g; 4 doses/day) or meropenem (2 g; 3 doses/day). If indications of sepsis emerged, glycopeptides were

further introduced into the treatment regimen. We regularly educated all caregivers (doctors and nurses) on duty at our alloHSCT unit to provide 2 g meropenem (3 doses/day) intravenously in case of neutropenic fever without delay or further instructions during night-duty. If treatment with meropenem was started, de-escalation to a less broad and more specific antibiotic according to clinical symptoms would be initiated the next day having banned the immediate danger of Gram-negative sepsis during neutropenia (Fig. 1).

Endpoints

The primary endpoints of the study were OS and NRM at 3 years after alloHSCT. Secondary endpoints were occurrence of BSI, days of fever and antibiotic treatment, antibiotic consumption, admission to an intensive care unit (ICU), neutropenic fever and newly detected MDRO colonization within 100 days after alloHSCT. Further secondary endpoints were aGVHD and cGVHD, viral and fungal infections, as well as duration of hospital stay.

Statistical analysis

All statistical analyses were performed using SPSS (version 28, IBM Corporation, SPSS Institut Inc. Chicago, IL), Graph Pad Prism (version 9.2.0. 283, GraphPad Software, San Diego, CA), BiAS (version 11.12, epsilon-Verlag GbR, Hochheim Darmstadt, HE) and Microsoft Excel (version 16.58, Microsoft Corporation, Redmond, WA). Categorical variables were captured as absolute count or percentage, continuous variables as mean (plus SD) or median. Pearson chi-squared test or Fisher exact test were used to compare categorical variables. Mann-Whitney U test or the Student t test were used to compare continuous variables. Survival curves were calculated by the Kaplan-Meier method compared by logrank test. To perform competing risk analyses Aalen-Johansen-Schätzer estimator and Goetghebeur-Ryan's modified log rank test were used to generate cumulative incidence curves. The cumulative incidence of BSI and RR was calculated with death as competing event. NRM cumulative incidence was estimated using relapse as competing event. Cumulative incidence of GVHD was generated with death or relapse as competing event. Multivariate analysis was carried out to assess independent risk factors (SAP, BSI, Age > 60 years, DRI, active disease, conditioning intensity: MAC vs. RIC) for NRM and OS using Cox proportional hazard. Consumption of antibacterial drugs is given in defined daily doses per 100 care days (DDD/100CD), the defined daily dose was chosen according to local treatment routine and Poisson-Test was performed to test for significant difference.

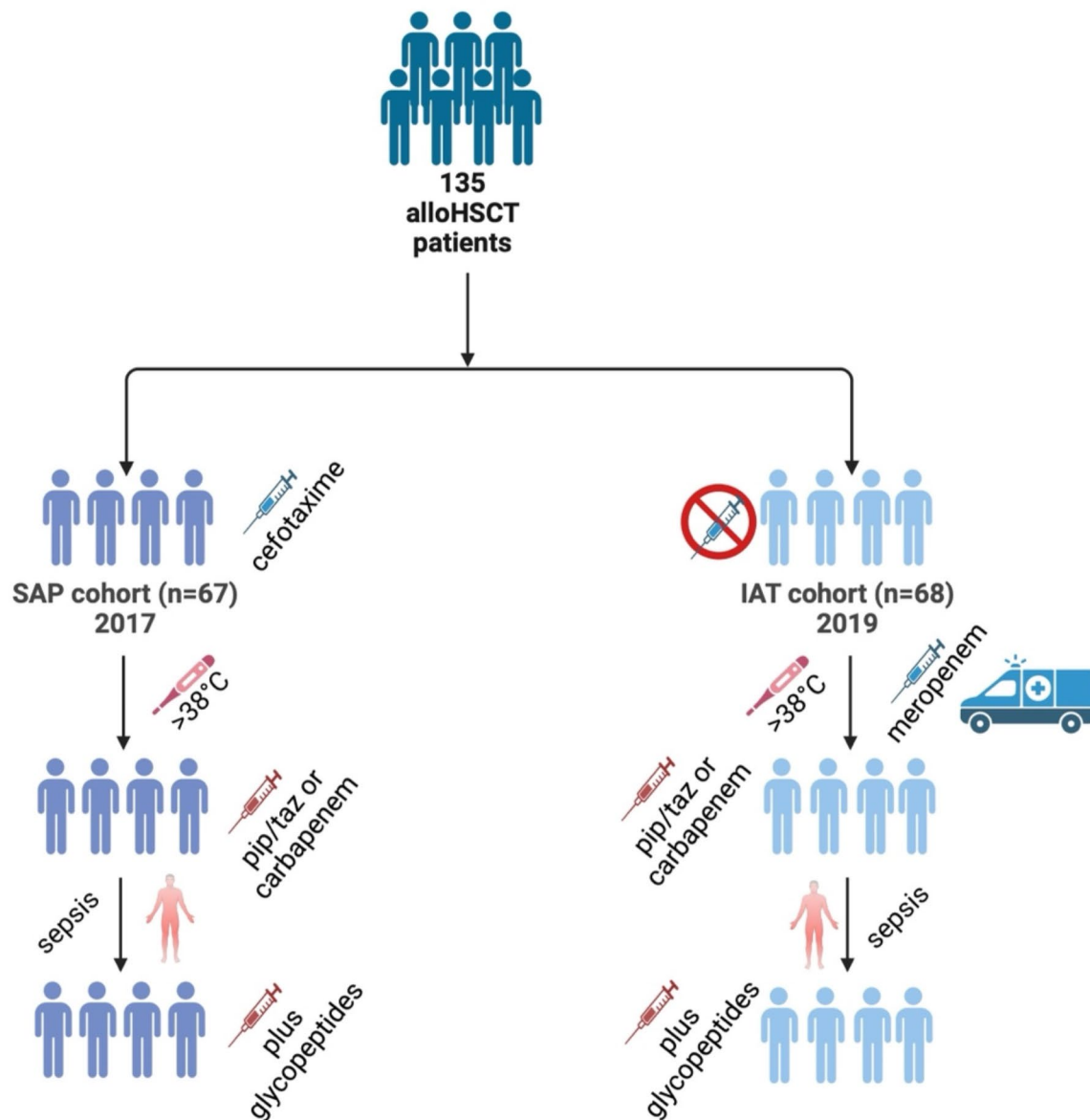


Fig. 1 Schematic overview of allogeneic stem cell transplantation (allo-HSCT) patient cohort and antibiotic treatment strategy for patient with systemic antibiotic prophylaxis (SAP) versus patients with an interventional antibiotic treatment (IAT), created with BioRender.com

Results

Patient characteristics

A total of 135 alloHSCT patients were included into the analysis with 67 patients in the SAP cohort and 68 patients in the IAT cohort (Tables 1 and Table 1 supplement). The most frequent hematologic diseases included acute myeloid leukemia ($n=68$, 50%), acute lymphoblastic leukemia ($n=32$, 24%), myelodysplastic syndromes ($n=10$, 7%) and primary myelofibrosis ($n=9$, 7%) which

were equally distributed between cohorts. The overall median age was 56 years (range 18–78 years). At the time of alloHSCT, a higher proportion of patients in the SAP cohort were in complete remission compared to those in the IAT cohort (66%, $n=44$ vs. 51%, $n=35$). Similarly, 34% ($n=23$) and 49% of patients ($n=33$) in the SAP and IAT cohorts, respectively, presented with active disease. Consequently, 49% of patients in the IAT cohort had a high-risk and very high-risk Disease Risk Index (DRI), while in the SAP cohort, the corresponding percentage was 41% ($p=ns$). SAP and IAT cohorts differed significantly

Table 1 Summary of patient characteristics

Patient Characteristics	all patients <i>n</i> = 135	SAP <i>n</i> = 67	IAT <i>n</i> = 68	<i>p</i> -value
Type of underlying disease: No. (%)				0.061
Acute myeloid leukemia (AML)	68 (50)	39 (58)	29 (43)	
Acute lymphatic leukemia (ALL)	32 (24)	19 (28)	13 (19)	
Myelodysplastic syndrome (MDS)	10 (7)	3 (5)	7 (10)	
Non-Hodgkin lymphomas (NHL)	6 (4)	0 (0)	6 (9)	
Others*	19 (15)	6 (9)	13 (19)	
Male sex: No (%)	85 (63)	42 (63)	43 (63)	> 0.999
Age at alloHSCT: median [range], y	56 [18–78]	56 [18–78]	55 [19–77]	0.143
Remission-status at alloHSCT: No. (%)				0.117
complete remission	79 (59)	44 (66)	35 (51)	
active disease	56 (41)	23 (34)	33 (49)	
Disease Risk Index (DRI): No. (%)				0.491
low/ intermediate risk	74 (55)	39 (59)	35 (51)	
high/ very high risk	61 (45)	28 (41)	33 (49)	
Conditioning MAC: No. (%)	66 (49)	40 (60)	26 (38)	0.025
Donor Type: No. (%)				0.178
MRD 10/10	23 (17)	16 (24)	7 (10)	
MUD 10/10	83 (61)	38 (57)	45 (66)	
MMUD (9/10 or 8/10)	11 (8)	4 (6)	7 (10)	
Haploidentical	18 (13)	9 (13)	9 (13)	
GvHD prophylaxis, No. (%)				0.279
ATG	89 (66)	41 (61)	48 (71)	
PTCY	25 (19)	13 (19)	12 (18)	0.828
CNI + MTX	26 (20)	14 (21)	12 (18)	0.668
CNI + MMF	109 (80)	53 (79)	56 (82)	
Time from diagnosis/relapse to alloHSCT: median [range], m	4 [0–19]	3.77 [0–19]	5 [0–19]	0.124
HCT-CI: No. (%)				0.025
low/ intermediate risk	73 (54)	43 (64)	30 (44)	
high risk	62 (46)	24 (36)	38 (56)	

Abbreviations: alloHSCT, allogeneic hematopoietic stem cell transplantation; ATG, antithymocyte globulin; CNI, calcineurin inhibitor; GvHD, graft-versus-host-disease; HCT-CI, hematopoietic stem cell transplantation comorbidity index; IAT, interventional antibiotic treatment; MAC, myeloablative conditioning MMUD, mismatched unrelated donor; MRD, matched related donor; MUD, matched unrelated donor; MTX, methotrexate; MMF, mycophenolate mofetil; m, months; PTCY, post transplant cyclophosphamide; SAP, systematic antibiotic prophylaxis; y, years; *hodgkin lymphomas, chronic myeloid leukemia, primary myelofibrosis, aplastic anaemia, chronic lymphatic leukemia, multiple myeloma, T cell prolymphocytic leukemia

P-values analyses differences between the SAP and the IAT-cohort, significance expected if $p < 0.05$

with respect to a high risk HCT-CI score ($n = 36$ patients, 24% vs. $n = 56$, 38%, $p = 0.025$) which is also reflected in a significantly more frequent use of MAC in the SAP than IAT cohort ($n = 40$ patients, 60% vs. $n = 26$ patients, 38%; $p = 0.025$). The median time from diagnosis or relapse to alloHSCT was 4 months (range 0–19 months), and peripheral blood stem cells were used in most patients (82%, $n = 111$) with a median CD34 positive cell dose of 6×10^6 per kg body weight (range $1.32\text{--}11.1 \times 10^6$ per kg body weight). The majority of patients (61%, $n = 83$) received a stem cell graft from a human leukocyte antigen (HLA)-matched unrelated donor (MUD), 17% ($n = 23$) from an HLA-matched related donor (MRD), 8% ($n = 11$) from a mismatched unrelated donor (MMUD) and 13% ($n = 18$) from an HLA-haploidentical donor. GVHD prophylaxis included ATG in 66% of patients ($n = 89$).

Transplantation-related characteristics

The median time to engraftment was 15 days (range, 3–130 days) for platelets and 17 days (range, 9–33 days) for neutrophils. During the immediate transplantation period (time from transplantation to discharge), patients stayed in hospital for a median of 24 days (range, 14–96 days) and 13 out of 135 patients (10%) were transferred to the ICU. After discharge, 45 patients (33%) required readmission within 100 days post-transplantation, 29 of these for infectious diseases (21% of all patients). Fungal and viral infections were distributed equally between groups. We observed a trend to more patients experiencing febrile episodes in the IAT cohort (62% of patients, $n = 42$) than in the SAP cohort (46%, $n = 31$; $p = 0.085$). The median number of febrile days

were 2 days (range 0–11 days) in both cohorts. If febrile days during ATG treatment were excluded, the number ranges between 0 and 10 days with a mean of 0 days (SAP cohort) and 1 day (IAT cohort) without significant difference. All transplant-related characteristics are depicted summarized in Table 2.

Replacement of SAP by IAT is safe and feasible in alloHSCT

We analyzed if the IAT strategy ultimately affected survival rates compared to SAP. At a median follow-up of 34.5 months, the median OS was not reached in both groups. OS at three years after alloHSCT was similar between the different antibiotic treatment strategies (SAP: 3y OS 68.9%, 95%CI 52–86%, IAT: 3y OS 66%, 95%CI 49–83%, $p=0.76$; Fig. 2a). At three years, 13 out of 135 patients had died from non-relapse mortality resulting in an incidence of 10% without differences between cohorts (SAP 3y NRM 9.75%, [95%CI 4.55–20.91%] vs. IAT 3y NRM 10.96% [95%CI 5.41–22.23%]), Fig. 2b. Causes of death included bacterial infections (pneumonia), viral infections (encephalitis), fungal infections (pneumonia), GVHD and therapy-associated toxicities. The CI of relapse at three years was 30% (41 of 135 patients) and did not differ significantly between the two cohorts (CI SAP-cohort 27% [95%CI 18.49–40.70%] and CI IAT-cohort 44% [95%CI 26.80–73.35%], $p=0.317$), Fig. 1 supplement.

We selected age, DRI und active disease status as variables for our multivariate analysis, recognizing them as established negative prognostic indicators for outcomes. Our findings identified age above 60 years at alloHSCT as an independent risk factor for fatal outcome in OS and

NRM (OS: HR 2.299, 95%CI 1.148–4.606, $p=0.019$ and NRM: HR 3.618, 95%CI 1.073–12.199, $p=0.038$, Table 3), whereas a high risk/very high risk DRI was associated with a fatal outcome only in OS (HR 1.945; 95%CI 1.000–3.785, $p=0.050$). In this study, neither the tested antibiotic intervention, the occurrence of BSI, nor the use of a MAC showed an impact on OS or NRM (Table 3 and Supplement Table 2).

IAT leads to a lower administration of antibiotics in alloHSCT

To investigate if IAT affected the consumption of antibiotics, we calculated the amount of antibiotics in DDD/100CD for each study cohort. This analysis revealed a lower administration of cefotaxime, teicoplanin, meropenem and vancomycin ($p<0.001$ for each drug, Fig. 3), while the consumption of piperacillin/tazobactam increased from 14.3 to 23.6 DDD/100CD ($p<0.001$). There were no differences in the administration of imipenem (7.6 versus 7.8 DDD/100CD), Fig. 3. Of note, IAT was associated with a median of 18 days (range 5–52 days) of systemic antibiotic treatment compared to 24 days (range 11–39 days) with SAP, resulting in a significantly lower patient antibiotic exposure ($p=0.0001$), Table 2.

IAT is associated with a higher incidence of BSI in alloHSCT

To test the impact of IAT on infection rates, we compared the incidence rates of BSIs between cohorts. Here, we found that the cumulative incidence of BSIs was three times higher as in the IAT compared to the SAP cohort within

Table 2 Summary of transplant-related characteristics

Transplantation-related characteristics	All patients <i>n</i> = 135	SAP <i>n</i> = 67	IAT <i>n</i> = 68	<i>p</i> -value
ANC recovery: median [range], d	17 [9–33]	17 [11–33]	16 [9–30]	0.149
PLT recovery: median [range], d	15 [3–130]	15 [3–130]	17 [10–46]	0.069
Hospital length of stay: median [range], d	24 [14–96]	25 [18–47]	22 [14–96]	0.101
Neutropenic fever: No. (%)	73 (54)	31 (46)	42 (62)	0.085
Febrile days: median [range], d	2 [0–11]	2 [0–11]	3 [0–9]	0.167
Febrile days without ATG: median [range], d	0 [0–10]	0 [0–10]	1 [0–9]	0.237
Days of antibiotics: median [range], d	20 [5–52]	24 [11–39]	18 [5–52]	0.001
CMV reactivation: No. (%)	49 (36)	26 (39)	23 (34)	0.594
Other viral infection during alloHSCT: No. (%)	30 (22)	12 (18)	18 (26)	0.301
All Fungal infection: No. (%)	17 (13)	8 (12)	9 (13)	> 0.999
Pulmonary fungal infection: No. (%)	14 (10)	7 (10)	7 (10)	> 0.999
ICU admission: No. (%)	13 (10)	4 (6)	9 (13)	0.243
Readmission: No. (%)	45 (33)	22 (33)	23 (34)	> 0.999
Readmission for infectious disease: No. (%)	29 (21)	12 (18)	17 (25)	0.403

Abbreviations: alloHSCT, allogeneic hematopoietic stem cell transplantation; ANC, absolute neutrophil count; d, days; CMV, cytomegalovirus; IAT, interventional antibiotic treatment; PLT, platelets; SAP, systematic antibiotic prophylaxis

P-values analyses differences between the SAP and the IAT-cohort, significance expected if $p<0.05$

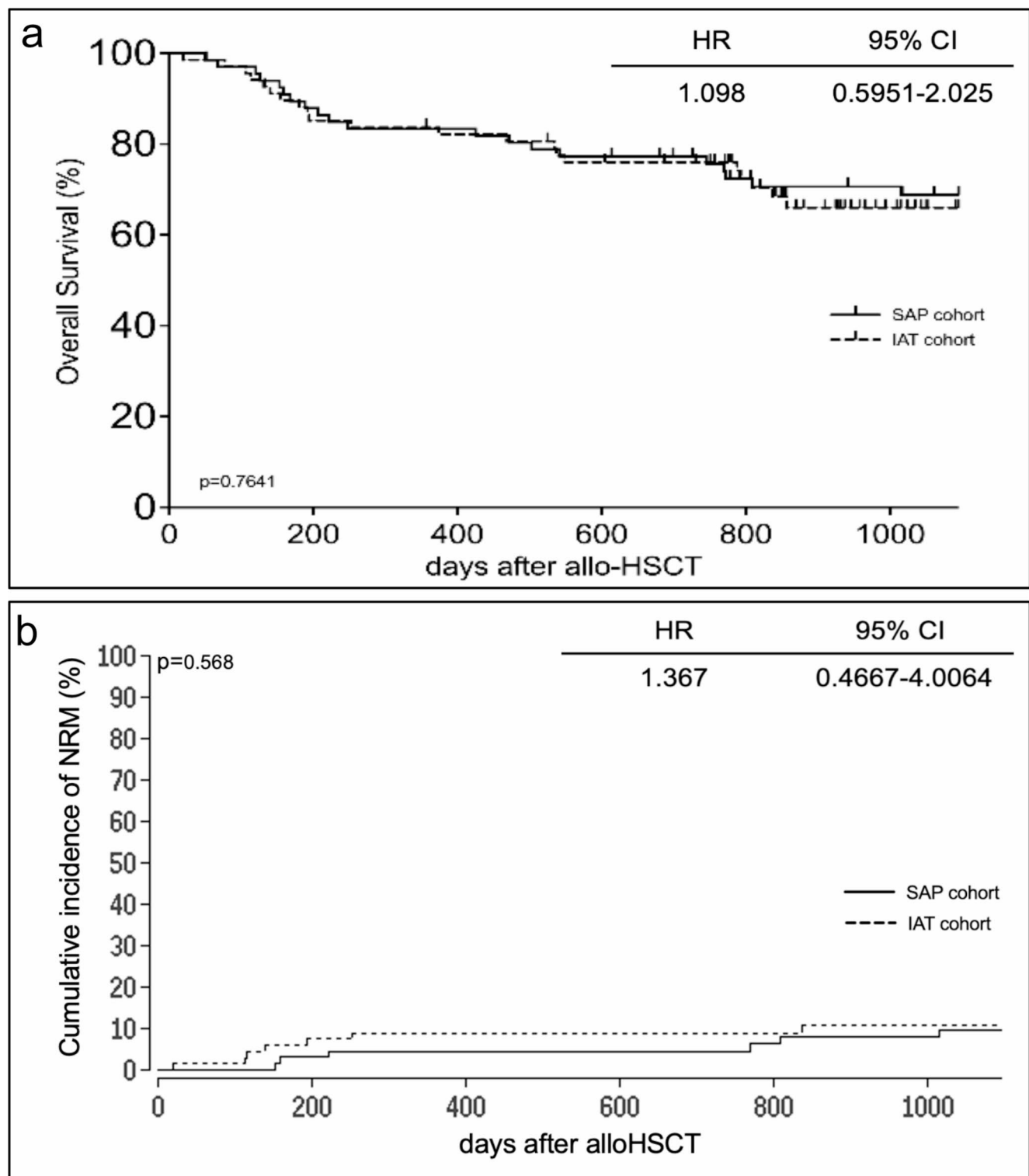


Fig. 2 Outcomes at three years after allogeneic stem cell transplantation (allo-HSCT) for patient with systemic antibiotic prophylaxis (SAP, solid line) versus patients with an interventional antibiotic treatment (IAT, dotted line). Kaplan-Meier curve illustrates overall sur-

vival (**a**), Cumulative incidence curves are presenting the non-relapse mortality (NRM) (**b**) and relapse rate (**c**). The competing risk analyses indicated the depicted Hazard Ratios (HR) with 95% confidence interval (CI) in the IAT versus SAP cohort

Table 3 Multivariate analysis of risk factors for death from any cause (above) and from non-relapse mortality (below)

Death from any cause			
Risk Factor	HR	95%CI	p-value
SAP	1.008	0.530–1.915	0.982
BSI	0.873	0.433–1.760	0.704
Age > 60 years	2.299	1.148–4.606	0.019
DRI: high/ very high risk (vs. favorable/ intermediate risk)	1.945	1.000–3.785	0.050
active disease (vs. CR)	1.308	0.650–2.630	0.452
MAC (vs. RIC)	0.959	0.443–2.075	0.916
Non-relapse Mortality			
Risk Factor	HR	95% CI	p-value
SAP	0.709	0.214–2.344	0.573
BSI	0.385	0.083–1.792	0.224
Age > 60 years	3.618	1.073–12.199	0.038
DRI: high/ very high risk (vs. favorable/ intermediate risk)	1.663	0.532–5.196	0.382
active disease (vs. CR)	0.677	0.190–2.407	0.546
MAC (vs. RIC)	0.739	0.190–2.875	0.662

Abbreviations: BSI blood stream infections, CR complete remission, HR hazard ratio, SAP systematic antibiotic prophylaxis, DRI disease risk index, 95%CI 95% confidence interval, MAC myeloablative conditioning, RIC reduced intensity conditioning

P-values analyses differences between the SAP and the IAT-cohort, significance expected if p

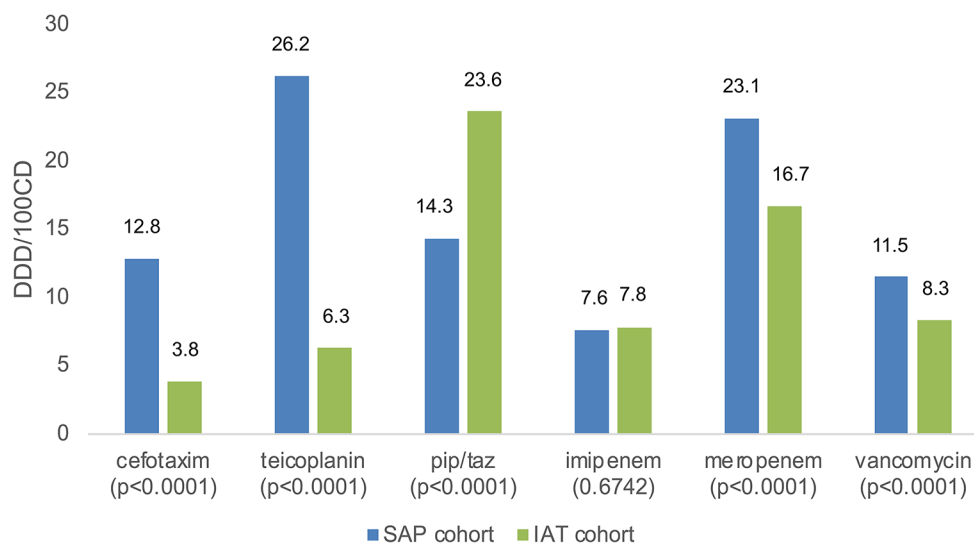
the first 100 days after allo HSCT (SAP: 13% (95%CI 7.3–24.7%) IAT: 40% (95%CI 29.6–53.2%) ($p < 0.001$), Fig. 4a. We further analyzed the pathogen spectrum causing the BSIs (Fig. 4b). In the IAT cohort we noted an increase in the presence of a wide range of bacteria, particular among fully susceptible species e.g., *Staphylococcus epidermidis* from 4 to 8 ($p = 0.365$) and of *Escherichia coli* from 0 to 9 ($p = 0.003$) BSI episodes in SAP versus IAT. In contrast, two BSI by MDRO, *Pseudomonas aeruginosa* classified as multi-resistant gram-negative (3MRGN, $n = 1$) and

vancomycin-resistant *Enterococcus faecium* (VRE, $n = 1$) could be detected in the IAT cohort. There was no significant difference between the two cohorts in terms of newly detected colonization with MDR bacteria during conditioning and after alloHSCT (VRE 10%; 3MRGN or 4MRGN 4%; methicillin-resistant *Staphylococcus aureus* (MRSA) 0%), however a tendency towards more *Enterobacteriaceae* with extended-spectrum β -lactamase ESBL in the SAP cohort was noticed (7% SAP cohort versus 1% IAT cohort; $p = 0.053$). In both cohorts the incidence of *Clostridium difficile* infections (CDI) within 100 days after alloHSCT was similar (12% IAT cohort vs. 11% SAP cohort) Supplementary Table 3.

IAT does not impact graft-versus-host disease development

Based on previous findings that a SAP-mediated disruption of the microbiome might impact the development of GVHD [22, 23], we tested the incidence rates of acute and chronic GVHD between cohorts. The CI of aGVHD grade II–IV at 100 days was 29.85% (95%CI: 20.68–43.09%) in the SAP cohort and 38.71% (95%CI: 28.64–52.33%) in the IAT cohort without significant difference ($p = 0.307$). cGVHD of any grade at 3 years occurred with a CI of 50.01% (95%CI 39.03–64.08%) in the SAP cohort and 44.75% (95% CI: 33.96–58.97%) in the IAT-cohort ($p = 0.952$) respectively, resulting in similar survival free of acute GvHD grade III–IV, chronic GVHD requiring systemic immunosuppressive therapy and relapse (GRFS) of the SAP-cohort (36.99%, 95%CI 19.88–54.1%) and the IAT cohort (16.89%, 95%CI 0–100%, $p = 0.282$). This holds also true for the CI of severe cGVHD at 3 years post-alloHSCT with 27.86% in the SAP cohort (95%CI 18.81–41.28%) versus 13.46% in the IAT cohort (95%CI 7.33–24.72%, $p = 0.098$) Fig. 5.

Fig. 3 Antibiotic consumption is compared between systematic antibiotic prophylaxis (SAP, blue) and interventional antibiotic treatment (IAT, green), stratified using the defined daily dose per 100 care days (DDD/100CD), p values were calculated according to poisson-test



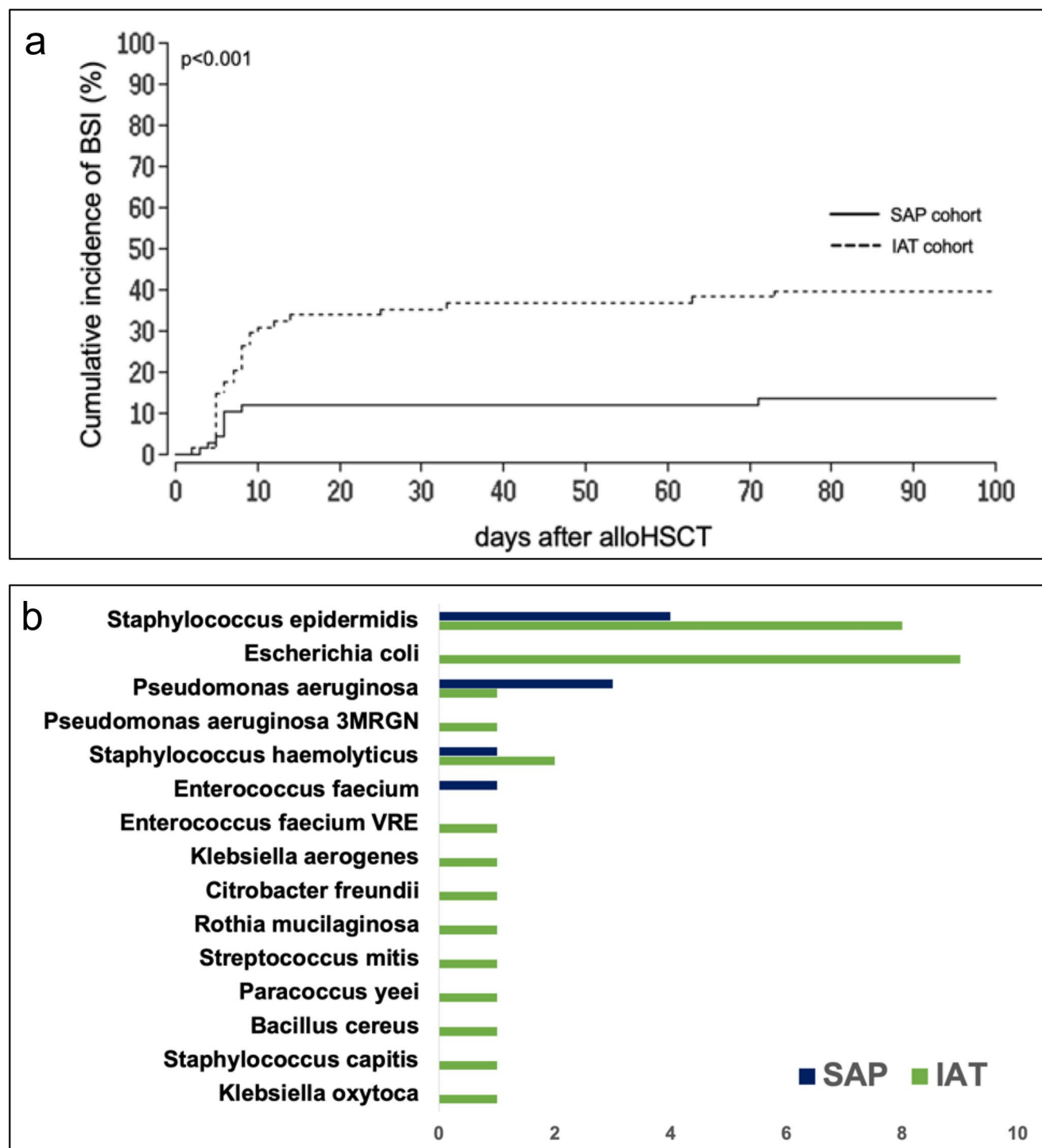


Fig. 4 Cumulative incidence curves are presenting occurrence of blood stream infections (BSI) within 100 days after alloHSCT of patients treated with a systematic antibiotic prophylaxis (SAP) versus an interventional antibiotic treatment (IAT). The competing risk analyses indicated the depicted Hazard Ratios (HR) with 95% confidence interval

(CI) in the IAT versus SAP cohort (**a**). A detailed analyses of different bacterial species causing blood stream infection detected in blood cultures drawn from patients treated with a SAP versus an IAT (**b**). 3MRGN indicating multi-resistant gram-negative (3 groups of specific antibiotics) and VRE, vancomycin-resistant Enterococcus

Discussion

In the field of alloHSCT and cellular therapies, patients continue to be at risk for post-transplantation infections due to delayed recovery of immunity and disturbances of mucosal barrier, despite evolving pre-transplantation therapies and underlying diseases profiles [24]. While 20th century guidelines recommended universal antibiotic treatment for fever and severe neutropenia [25, 26], recent recommendations advocate for a more nuanced approach, evaluating clinical likelihood of infections and excluding confounding causes of fever e.g. cytokine release syndrome (CRS) before initiating antibiotics [27, 28].

In this retrospective study, we analyzed our single-center experience with alloHSCT patients without antibiotic prophylaxis but with stringent guidelines for prompt antibiotic intervention. Our findings demonstrate the safety of this strategy, with no adverse impact on OS or NRM, despite a higher BSI incidence. Of note, IAT significantly shortened antibiotic treatment duration and reduced antibiotic consumption, including not only cefotaxime but also broad spectrum carbapenem and glycopeptide antibiotics. However, larger patient numbers are necessary to confirm a potential benefit on GVHD.

The use of SAP during neutropenia has been standard of care in many alloHSCT centers, based on studies published

between 1973 and 2005 indicating lower infection rates, bacteraemia and fever episodes without impacting mortality [5, 29]. While fluoroquinolones like levofloxacin or ciprofloxacin were commonly used, third generation cephalosporins (e.g. cefotaxim) served as alternatives in case of intolerances, allergies or drug interactions [30]. However, adverse events and the emergence of resistant bacteria associated with levofloxacin prompted a shift towards the use of cefotaxime as SAP in our center [5, 31]. Despite previous studies, our data did not show a significant benefit of SAP on OS, whether 100 days or 3 years after alloHSCT. The potential advantages (non-significant better OS and lower ICU admission rate) of the SAP cohort were offset by higher comorbidity indexes and lower CR rates in the IAT group, warranting further studies to fully elucidate this question.

Contrary to the assumption that SAP reduces mortality by lowering BSI rates, our results emphasize the critical role of timely antibiotic administration in managing neutropenic fever [32]. Our IAT strategy involving immediate meropenem administration (by nursing staff) upon neutropenic fever [33] onset followed by rapid de-escalation of this broad-spectrum empiric therapy did not result in increased meropenem consumption. In line with current guidelines we would de-escalate to piperacillin/tazobactam in most of the cases [33], which is also reflected in the increase of piperacillin/tazobactam consumption in the IAT cohort. Given

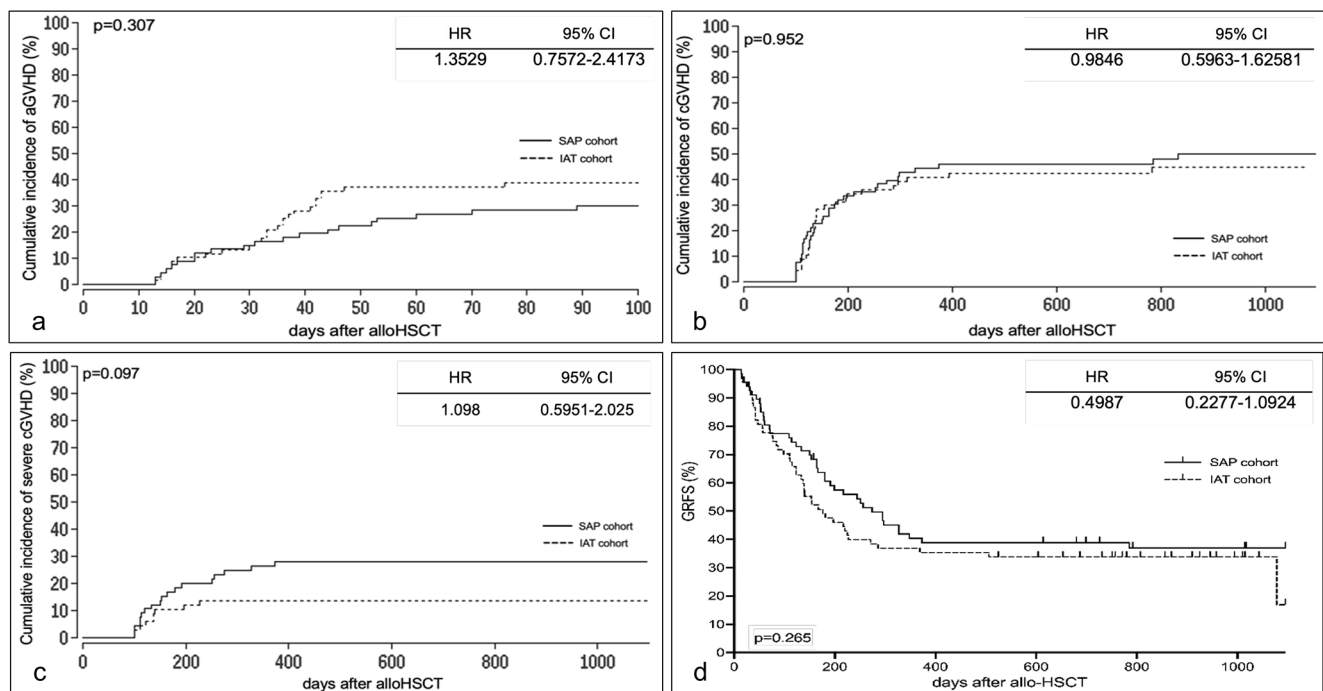


Fig. 5 Cumulative incidence curves are presenting occurrence of acute graft versus host disease (aGVHD) grade II-IV (a), chronic graft versus host disease (cGVHD) of any grade (b) and severe cGVHD (c) at 3 years after alloHSCT of patients treated with a systematic antibiotic prophylaxis (SAP) versus an interventional antibiotic treatment (IAT).

Survival free of acute GVHD grade III-IV, chronic GVHD requiring systemic immunosuppressive therapy and relapse-free survival (GRFS) is depicted in Fig. 5d. The competing risk analyses indicated the depicted Hazard Ratios (HR) with 95% confidence interval (CI) in the IAT versus SAP cohort

that microbial analyses for bacteremia are most accurate when the blood is taken before the first administration of antibiotics (without SAP in place), anti-infective therapy could then be tailored towards the susceptibility of the identified bacteria [34]. Furthermore, bacterial infections potentially prevented by SAP (e.g., ciprofloxacin) are typically caused by susceptible bacteria that can be treated sufficiently by empirical antibiotics. In contrast, bacteremia-associated mortality is mostly caused by MDR pathogens, which are unlikely to be prevented by SAP [32, 35].

Despite advancements in alloHSCT, GVHD remains a significant concern, contributing to substantial morbidity and mortality [36]. The disruption of the human microbiota caused by alloHSCT seems to be part of the complex pathophysiology of GVHD [37], with antibiotic usage contributing to bacterial dysbiosis [38], development of GVHD and poor outcomes after alloHSCT [39–41]. While the association between capanems (e.g. Meropenem and Imipenem) and piperacillin/tazobactam with severe perturbation of gut microbial compositions and an increased GVHD-related mortality has been documented [38, 41, 42], the use of distinct oral antibiotics for gut decontamination like rifaximin, has shown promising results in reducing GVHD incidence and improving outcome [23]. Although our study did not reveal significant differences in GVHD incidence between the two groups, the observed decrease in antibiotic treatment days within the IAT cohort might be beneficial for microbiota diversity and GVHD development [40]. Larger patient numbers and detailed microbiome analyses are needed to confirm a possible benefit of IAT on GVHD development.

While our study sheds light on the potential benefits of replacing SAP with IAT post-alloHSCT, certain limitations must be acknowledged. This retrospective analysis comprises a relatively small sample size, and other factors (e.g., supportive treatment or prior antibiotic treatment), may have influenced observed outcomes. Additionally, deviations from recommended SAP strategies, such as our use of cefotaxim instead of a fluorochinolone, may limit the ability to generalize conclusions. It is also important to note that the strong anti-anaerobic antibiotics used in the IAT cohort may have compensated for the lack of prophylaxis, a limitation we cannot fully address. Follow-up studies, such as a randomized trial comparing narrower-spectrum antibiotics like cefepime, without prophylaxis, against standard prophylaxis, could provide further insights.

Based on our single center experience, we suggest that replacing SAP with IAT after alloHSCT is safe and effective, with possible benefits including reduced antibiotic consumption. However, prospective studies with larger patient cohort are warranted to validate these findings and refine alloHSCT infection management guidelines.

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Declarations

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